

Boss Challenge — Paper 40 (Class 7)

Weather, Climate & Adaptations | Transportation in Animals & Plants | Data Handling | Rational Numbers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The day-to-day state of the air at a place - its heat, rain and wind on that very day - is called the:
(A) climate
(B) season
(C) atmosphere
(D) weather
2. The red fluid that carries digested food, oxygen and wastes to every part of our body is the:
(A) saliva
(B) sweat
(C) blood
(D) urine
3. Facts or figures collected together, such as a week's recorded temperatures, are together called:
(A) a survey
(B) an average
(C) a graph
(D) data
4. Any value that can be expressed as p/q , with p and q integers and q not equal to zero, is termed a:
(A) rational number
(B) a whole number only
(C) a number that cannot be a fraction
(D) an odd number
5. The average pattern of the weather of a place, worked out over a long period of many years, is its:
(A) humidity
(B) forecast
(C) climate
(D) weather
6. The red colour of our blood comes from a special pigment carried inside the red blood cells, called:
(A) platelet
(B) haemoglobin
(C) plasma
(D) chlorophyll

7. Adding up all the observations and dividing by how many there are gives the:
- (A) median
 - (B) mean (average)
 - (C) mode
 - (D) range
8. In the rational number $\frac{3}{7}$, the number 7 written below the line is called the:
- (A) quotient
 - (B) denominator
 - (C) numerator
 - (D) remainder
9. At a weather station, the instrument used to measure how hot or cold the air is, is the:
- (A) thermometer
 - (B) wind vane
 - (C) rain gauge
 - (D) barometer
10. The pale yellow liquid part of blood, in which the blood cells float and are carried along, is the:
- (A) platelets
 - (B) marrow
 - (C) haemoglobin
 - (D) plasma
11. Over 5 days the maximum temperatures were 30, 32, 31, 29 and 33 degrees C. Their mean temperature is:
- (A) 31 degrees C
 - (B) 155 degrees C
 - (C) 29 degrees C
 - (D) 33 degrees C
12. Every integer is also a rational number, because any integer can be written as a fraction with denominator:
- (A) 0
 - (B) 1
 - (C) 2
 - (D) the integer itself
13. Rainfall at a place is collected and measured in millimetres using a:
- (A) thermometer
 - (B) rain gauge
 - (C) anemometer
 - (D) hygrometer

14. The tiny blood cells that help blood clot and seal a cut so bleeding stops are the:
- (A) nerve cells
 - (B) platelets
 - (C) white blood cells
 - (D) red blood cells
15. In a set of observations, the value that occurs most often is called the:
- (A) median
 - (B) mean
 - (C) mode
 - (D) range
16. The standard (lowest) form of the rational number $\frac{6}{8}$ is:
- (A) $\frac{12}{16}$
 - (B) $\frac{3}{4}$
 - (C) $\frac{6}{8}$
 - (D) $\frac{2}{4}$
17. The maximum and minimum readings noted each day at a weather station describe one element of weather, namely the:
- (A) rainfall
 - (B) wind speed
 - (C) temperature
 - (D) humidity
18. The muscular organ that keeps pumping to push blood around the whole body is the:
- (A) kidney
 - (B) lung
 - (C) stomach
 - (D) heart
19. On a week's rainfall record the value 0 mm appears on 4 of the 7 days, more often than any other. The mode of the rainfall is:
- (A) 0 mm
 - (B) the highest rainfall
 - (C) 4 mm
 - (D) 7 mm
20. Which is the greater rational number: $-\frac{3}{4}$ or $-\frac{1}{4}$?
- (A) neither, both are zero
 - (B) $-\frac{1}{4}$
 - (C) they are equal
 - (D) $-\frac{3}{4}$

21. Polar bears survive the freezing Arctic chiefly because of a thick layer of fat and dense fur, which work as:
- (A) insulation against the cold
 - (B) camouflage from prey
 - (C) a store of water
 - (D) better eyesight
22. The 'lub-dub' sound that a doctor hears through a stethoscope on the chest is produced by the:
- (A) air rushing into the lungs
 - (B) food moving in the stomach
 - (C) blood being made in bones
 - (D) closing of the heart valves
23. When all the observations are arranged in order, the value that lies right in the middle is the:
- (A) median
 - (B) mode
 - (C) mean
 - (D) range
24. On the number line, the rational number $-\frac{2}{3}$ lies between:
- (A) -2 and -1
 - (B) 0 and 1
 - (C) 1 and 2
 - (D) -1 and 0
25. The polar regions stay bitterly cold all year mainly because the Sun's rays reach them very:
- (A) directly from overhead
 - (B) only at night
 - (C) after passing through water
 - (D) slantingly, so they spread out
26. Blood vessels that carry blood away from the heart to the rest of the body are the:
- (A) nerves
 - (B) arteries
 - (C) veins
 - (D) capillaries
27. The five temperatures 28, 30, 31, 33, 35 degrees C are already in order. Their median is:
- (A) 32 degrees C
 - (B) 31 degrees C
 - (C) 28 degrees C
 - (D) 35 degrees C

- 28.** The additive inverse (the opposite) of the rational number $\frac{5}{9}$ is:
- (A) $-\frac{5}{9}$
 - (B) $\frac{9}{5}$
 - (C) 0
 - (D) $\frac{5}{9}$
- 29.** A region near the equator that stays hot and receives heavy rain almost all year is a:
- (A) tropical rainforest
 - (B) grassland
 - (C) polar region
 - (D) desert
- 30.** Blood vessels that carry blood from the body back towards the heart are the:
- (A) arteries
 - (B) windpipe
 - (C) capillaries
 - (D) veins
- 31.** The difference between the highest and the lowest values in a set of data is called the:
- (A) median
 - (B) range
 - (C) mode
 - (D) mean
- 32.** A winter morning temperature of -8 degrees C rises by 3 degrees C. The new reading, as a rational number, is:
- (A) -5 degrees C
 - (B) 5 degrees C
 - (C) 11 degrees C
 - (D) -11 degrees C
- 33.** Animals of the tropical rainforest, such as monkeys and toucans, are adapted to a climate that is:
- (A) hot and humid all year
 - (B) hot and dry
 - (C) cold and dry
 - (D) cool with snow
- 34.** Arteries and veins are linked by extremely thin-walled vessels where food and oxygen pass to the cells; these are the:
- (A) capillaries
 - (B) valves
 - (C) veins
 - (D) arteries

35. On one day the maximum temperature was 36 degrees C and the minimum was 24 degrees C. The temperature range that day was:
- (A) 12 degrees C
 - (B) 36 degrees C
 - (C) 60 degrees C
 - (D) 24 degrees C
36. The sum $\frac{1}{2} + \frac{1}{3}$ equals:
- (A) $\frac{1}{5}$
 - (B) $\frac{1}{6}$
 - (C) $\frac{5}{6}$
 - (D) $\frac{2}{5}$
37. Penguins of the Antarctic often crowd together into a tight huddle mainly to:
- (A) see predators better
 - (B) lay more eggs
 - (C) reduce heat loss and stay warm
 - (D) catch more fish
38. How many times the heart beats during one full minute is known as the:
- (A) body temperature
 - (B) heart rate (pulse rate)
 - (C) breathing depth
 - (D) blood pressure
39. A bar graph shows data using bars that all have the same width but different:
- (A) colours
 - (B) widths
 - (C) shapes
 - (D) heights (lengths)
40. The value of $\frac{3}{4} - \frac{1}{2}$ is:
- (A) $\frac{2}{2}$
 - (B) $\frac{1}{2}$
 - (C) $\frac{1}{4}$
 - (D) $\frac{3}{8}$
41. The streamlined body and webbed feet of a penguin are adaptations that help it mainly to:
- (A) dig deep burrows
 - (B) fly high in the air
 - (C) climb tall trees
 - (D) swim well in water

42. We can feel a pulse at the wrist because, with every heartbeat, an artery there:
- (A) throbs as blood is pushed through it
 - (B) fills with air
 - (C) stops moving briefly
 - (D) turns blue
43. A pictograph shows the collected data chiefly by means of:
- (A) straight lines only
 - (B) a single number
 - (C) pictures or symbols
 - (D) only words
44. The product $\frac{2}{3} \times \frac{6}{5}$ equals (in lowest form):
- (A) $\frac{12}{15}$
 - (B) $\frac{4}{5}$
 - (C) $\frac{8}{15}$
 - (D) $\frac{2}{5}$
45. The very large beak of a toucan helps it reach distant fruit and also helps the bird to:
- (A) dig into the soil
 - (B) store food for winter
 - (C) lose extra body heat
 - (D) make a loud warning light
46. The pair of bean-shaped organs that filter wastes out of the blood and make urine are the:
- (A) liver
 - (B) heart
 - (C) kidneys
 - (D) lungs
47. On a pictograph one bird-symbol stands for 10 birds. Four such symbols therefore stand for:
- (A) 40 birds
 - (B) 14 birds
 - (C) 10 birds
 - (D) 4 birds
48. The reciprocal of the rational number $\frac{4}{7}$ is:
- (A) $\frac{4}{7}$
 - (B) $\frac{7}{4}$
 - (C) $\frac{1}{4}$
 - (D) $-\frac{4}{7}$
49. The thick white fur of a polar bear, besides keeping it warm, also helps the bear by acting as:
- (A) a way to fly
 - (B) a swimming float
 - (C) a source of food
 - (D) camouflage against the snow

50. Plants take in water and dissolved minerals from the soil mainly through their:

- (A) fruits
- (B) roots
- (C) leaves
- (D) flowers

51. When a fair coin is tossed once, the chance (probability) that it shows heads is:

- (A) 2
- (B) 1
- (C) $\frac{1}{2}$
- (D) 0

52. The value of $\frac{5}{6}$ divided by $\frac{1}{2}$ is:

- (A) $\frac{5}{12}$
- (B) $\frac{1}{2}$
- (C) $\frac{5}{3}$
- (D) $\frac{10}{6}$

53. Some birds fly thousands of kilometres each year to escape harsh weather and find food; this seasonal journey is called:

- (A) hibernation
- (B) camouflage
- (C) migration
- (D) adaptation

54. The tube-like tissue that carries water and minerals upward from the roots to the leaves is the:

- (A) xylem
- (B) phloem
- (C) stomata
- (D) root hair

55. An event that is absolutely certain to happen has a probability of:

- (A) $\frac{1}{2}$
- (B) 100
- (C) 1
- (D) 0

56. Between any two different rational numbers, how many other rational numbers can you find?

- (A) none
- (B) exactly ten
- (C) exactly one
- (D) infinitely many

57. Weather can change within a single day, but to describe the climate of a place we average its weather over at least about:
- (A) twenty-five years
 - (B) six months
 - (C) twenty-five hours
 - (D) two or three days
58. The tissue that carries food made in the leaves to all other parts of the plant is the:
- (A) phloem
 - (B) xylem
 - (C) bark
 - (D) stomata
59. A patient's heart-rate readings over five checks were 70, 72, 74, 72 and 72 beats per minute. The mode of these readings is:
- (A) 70
 - (B) 360
 - (C) 74
 - (D) 72
60. If plasma makes up $\frac{11}{20}$ of a sample of blood, then the blood cells (the rest) make up:
- (A) $\frac{9}{20}$
 - (B) $\frac{31}{20}$
 - (C) $\frac{1}{20}$
 - (D) $\frac{11}{20}$
61. On a normal sunny day, the maximum temperature is usually recorded in the:
- (A) midnight
 - (B) early morning
 - (C) afternoon
 - (D) just before sunrise
62. The loss of water as vapour from the surface of a plant's leaves is called:
- (A) transpiration
 - (B) germination
 - (C) respiration
 - (D) digestion
63. On a bar graph of a week's rainfall, the tallest bar represents the day that had the:
- (A) highest temperature
 - (B) least rainfall
 - (C) most sunshine
 - (D) most rainfall

64. The rational number lying exactly halfway between 0 and 1 is:
- (A) 0
 - (B) 1
 - (C) 2
 - (D) $\frac{1}{2}$
65. Humidity, an element of weather, tells us the amount present in the air of:
- (A) dust
 - (B) water vapour
 - (C) smoke
 - (D) oxygen
66. Transpiration is useful to a tall tree because the water escaping from the leaves helps to:
- (A) pull more water up from the roots
 - (B) store water in the stem
 - (C) push food down to the roots
 - (D) make the leaves heavier
67. Find the average of these five whole numbers: 0, 1, 2, 3 and 4. It is:
- (A) 2
 - (B) 5
 - (C) 4
 - (D) 10
68. Written in its lowest terms, the rational number $-\frac{6}{10}$ is:
- (A) $-\frac{3}{10}$
 - (B) $\frac{3}{5}$
 - (C) $-\frac{3}{5}$
 - (D) $-\frac{6}{10}$
69. Elephants living in hot regions have very large ears that they flap mainly to help them:
- (A) store extra water
 - (B) hear very distant sounds
 - (C) throw off body heat and cool down
 - (D) scare away lions
70. Sweating in humans removes a little water and salt from the body and, importantly, also helps to:
- (A) add oxygen to the blood
 - (B) cool the body down
 - (C) warm the body up
 - (D) digest the food faster
71. Five pulse readings, arranged in order, are 68, 72, 72, 76 and 80. The median (middle value) is:
- (A) 80
 - (B) 68
 - (C) 76
 - (D) 72

72. The mixed number $2\frac{1}{2}$ written as an improper fraction is:

- (A) $\frac{2}{2}$
- (B) $\frac{4}{2}$
- (C) $\frac{5}{2}$
- (D) $\frac{3}{2}$

73. An animal that stays active mainly at night and rests by day to avoid the daytime heat is said to be:

- (A) migratory
- (B) herbivorous
- (C) aquatic
- (D) nocturnal

74. A tiny single-celled animal such as Amoeba has no blood or heart because it exchanges gases and food:

- (A) by pumping with a small heart
- (B) through a long network of veins
- (C) using green chlorophyll
- (D) directly through its surface by diffusion

75. Double bar graphs are especially useful because they let you:

- (A) hide one set of data
- (B) replace all numbers with one bar
- (C) compare two sets of data side by side
- (D) show only a single value

76. The product of any rational number and its reciprocal is always:

- (A) 1
- (B) the number itself
- (C) 0
- (D) its additive inverse

77. Both the Arctic in the north and the Antarctic in the south share a climate that is:

- (A) mild and pleasant
- (B) dry and very hot
- (C) extremely cold and windy
- (D) hot and rainy

78. After delivering oxygen to the body, the dark, oxygen-poor blood returns to the heart and is then sent to the:

- (A) kidneys to add water
- (B) skin to pick up sunlight
- (C) lungs to pick up oxygen
- (D) stomach to pick up food

79. An ordinary six-faced die is rolled once. The probability of getting a number greater than 6 is:
- (A) 0
 - (B) $\frac{1}{6}$
 - (C) 1
 - (D) 6
80. The decimal 0.75 written as a rational number in lowest terms is:
- (A) $\frac{7}{5}$
 - (B) $\frac{75}{100}$
 - (C) $\frac{3}{4}$
 - (D) $\frac{3}{5}$
81. Which everyday choice depends most directly on the weather of that particular day?
- (A) your house number
 - (B) the clothes you decide to wear
 - (C) your date of birth
 - (D) the language you speak
82. Oxygen-rich blood flowing out to the body is usually:
- (A) green
 - (B) dark bluish
 - (C) colourless
 - (D) bright red
83. To find the mean monthly rainfall over a full year, you divide the year's total rainfall by:
- (A) 7
 - (B) 1
 - (C) 365
 - (D) 12
84. The temperature falls from 2 degrees C to -3 degrees C. The change in temperature is:
- (A) a drop of 5 degrees C
 - (B) a drop of 1 degree C
 - (C) no change
 - (D) a rise of 5 degrees C
85. A sandy desert can be scorching by day yet cold at night mainly because the dry sand and air:
- (A) block the Sun completely
 - (B) heat up and cool down very fast
 - (C) trap heat like a thick blanket
 - (D) make their own heat
86. Into how many separate chambers is the human heart divided?
- (A) two
 - (B) three
 - (C) one
 - (D) four

87. The sum of six temperature readings is 180 degrees C. Their mean is found by computing:

- (A) $180 - 6 = 174$
- (B) $180 / 6 = 30$ degrees C
- (C) $180 + 6 = 186$
- (D) $180 \times 6 = 1080$

88. The numbers $-7/8$ and $7/8$ are equal in size but opposite in:

- (A) denominator
- (B) size
- (C) numerator
- (D) sign

89. The single great source of energy that drives nearly all the changes we group together as weather is the:

- (A) the rivers
- (B) the Moon
- (C) Sun
- (D) the soil

90. Which tissue carries water upward in a plant - xylem or phloem?

- (A) both equally
- (B) xylem
- (C) phloem
- (D) neither

91. Arranging a list of marks from lowest to highest is the natural first step when you want to find the:

- (A) median
- (B) range
- (C) mode
- (D) mean

92. To compare the rational numbers $2/3$ and $3/4$ fairly, the useful first step is to make their:

- (A) numerators equal
- (B) both into whole numbers
- (C) denominators equal
- (D) signs opposite

93. Two animals well adapted to life in a tropical rainforest are the:

- (A) polar bear and the penguin
- (B) toucan and the monkey
- (C) camel and the desert fox
- (D) yak and the snow leopard

- 94.** On a hot, dry, windy day a plant loses water by transpiration:
- (A) slower than on a cool day
 - (B) at exactly the same rate always
 - (C) only at night
 - (D) faster than on a cool, calm day
- 95.** A class recorded each student's favourite season. The season chosen by the greatest number of students is the:
- (A) median of the data
 - (B) mode of the data
 - (C) range of the data
 - (D) mean of the data
- 96.** Written in standard (lowest) form, the rational number $15/25$ is:
- (A) $15/25$
 - (B) $3/5$
 - (C) $5/3$
 - (D) $3/25$
- 97.** Camouflage helps an animal survive mainly by letting it:
- (A) make loud warning calls
 - (B) run much faster than others
 - (C) blend with its surroundings to avoid being seen
 - (D) store food for the winter
- 98.** The blood cells that defend the body by fighting germs and clearing infections are the:
- (A) plasma
 - (B) red blood cells
 - (C) platelets
 - (D) white blood cells
- 99.** If the mean of four numbers is 25, then the total (sum) of those four numbers must be:
- (A) 4
 - (B) 29
 - (C) 25
 - (D) 100
- 100.** The number 0 is a rational number because it can be written in the form p/q as:
- (A) $1/1$
 - (B) $0/1$
 - (C) $0/0$
 - (D) $1/0$